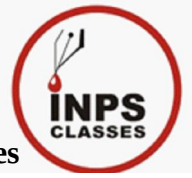


QUADRATIC EQUATION

DPP 03 Relation Between Roots & Coefficients



Date: 18.08.2026

Max Time: 60 Minutes

01. Let $f(x) = ax^2 + bx + c$, $a \neq 0$ and $\Delta = b^2 - 4ac$. If $\alpha + \beta$, $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$ are in GP, then
 (a) $\Delta \neq 0$ (b) $b\Delta = 0$ (c) $c\Delta = 0$ (d) $bc \neq 0$
02. Let α and β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}$, 2β be the roots of the equation $x^2 - qx + r = 0$. Then, the value of r is
 (a) $\frac{2}{9}(p-q)(2q-p)$ (b) $\frac{2}{9}(q-p)(2p-q)$ (c) $\frac{2}{9}(q-2p)(2q-p)$ (d) $\frac{2}{9}(2p-q)(2q-p)$
03. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers, satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is
 (a) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (b) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (c) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (d) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 + q) = 0$
04. Let k be a real number such that $k \neq 0$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -2k$ and $\alpha^2 + \beta^2 = 4k^2 - 2k$, then a quadratic equation having $\frac{\alpha + \beta}{\alpha}$ and $\frac{\alpha + \beta}{\beta}$ as its roots, equals
 (a) $4x^2 - kx + k = 0$ (b) $x^2 - 4kx + 4k = 0$ (c) $4x^2 - 4k + k = 0$ (d) $4kx^2 - 4kx + 1 = 0$
05. The equation, whose roots are the square of the roots of the equation $x^2 + bx + c = 0$, is
 (a) $a^2x^2 + b^2x + c^2 = 0$ (b) $a^2x^2 - (b^2 - 4ac)x + c^2 = 0$
 (c) $a^2x^2 - (b^2 - 2ac)x + c^2 = 0$ (d) $a^2x^2 + (b^2 - ac)x + c^2 = 0$
06. If one root of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is 1. Then, its other root is
 (a) $\frac{a(b-c)}{b(c-a)}$ (b) $\frac{c(a-b)}{a(b-c)}$ (c) $\frac{b(c-a)}{a(b-c)}$ (d) None of these
07. If difference of roots of the equation $x^2 - px + q = 0$ is 1, then $p^2 + 4q^2$ equals
 (a) $2q + 3$ (b) $(1 - 2q)^2$ (c) $(1 + 2q)^2$ (d) $2q - 3$
08. If α, β be the roots of $x^2 + px + q = 0$ and γ, δ be the roots of $x^2 + px + r = 0$, then $\frac{(\alpha - \gamma)(\alpha - \delta)}{(\beta - \gamma)(\beta - \delta)}$ is equal to
 (a) 1 (b) q (c) r (d) $q + r$

09. If product of the roots of the equation $x^2 - 3kx + 2e^{\sin k} - 1 = 0$ is 7, then its roots will be real, if
- (a) $|k| \leq 22\sqrt{\frac{7}{9}}$ (b) $|k| \leq 2\sqrt{\frac{7}{9}}$ (c) $|k| > 2\sqrt{\frac{7}{9}}$ (d) None of these
10. The expression $x^2 + 2(a + b + c)x + 3(bc + ca + ab)$ will be a perfect square, if
- (a) $a + b + c = 0$ (b) $ab + bc + ca = 0$ (c) $a = b = c$ (d) None of these
11. If p, q and r are positive and are in AP, then roots of the equation $px^2 + qx + r = 0$ are real, if
- (a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$ (b) $\left|\frac{p}{r} - 7\right| < 4\sqrt{3}$ (c) for all values of p, r (d) for no value of p, r
12. If one root of the equation $x^2 - 30x + p = 0$ is square of the other, then p is equal to
- (a) 120, 125 (b) 125, -216 (c) 125, 215 (d) None of these
13. Let α and β be the roots of $x^2 - x - p = 0$ and γ, δ be the roots of $x^2 - 4x + q = 0$. If α, β, γ and δ are in GP, then the integral values of p and q respectively, are
- (a) -2, -32 (b) -2, 3 (c) -6, 3 (d) -6, -32
14. If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the square of the reciprocals, then $\frac{a}{c}, \frac{b}{a}$ and $\frac{c}{b}$ are in
- (a) GP (b) HP (c) AP (d) None of these
15. If α and β are the roots of equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009}$ is equal to
- (a) -2 (b) -1 (c) 1 (d) 2
16. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. The equation, whose roots are α^{19} and β^7 , is
- (a) $x^2 - x - 1 = 0$ (b) $x^2 + x + 1 = 0$ (c) $x^2 + x - 1 = 0$ (d) $x^2 - x + 1 = 0$
17. Ramesh and Mahesh solve a quadratic equation. Ramesh reads its constant term wrongly and finds its roots as 8 and 2, whereas Mahesh reads the coefficient of x wrongly and finds its roots as 11 and -1. The correct roots of the equation are
- (a) 11, 1 (b) -11, 1 (c) 11, -1 (d) None of these
18. Let α and β be the roots of equation $px^2 + qx + r = 0, p \neq 0$. If p, q and r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is
- (a) $\frac{\sqrt{61}}{9}$ (b) $\frac{2\sqrt{17}}{9}$ (c) $\frac{34}{9}$ (d) $\frac{2\sqrt{13}}{9}$
19. The equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ has
- (a) infinite number of real roots (b) no real root
(c) exactly one real root (d) exactly four real roots
20. Let two numbers have arithmetic mean 9 and geometric mean 4. Then, these numbers are the roots of the quadratic equation
- (a) $x^2 + 18x + 16 = 0$ (b) $x^2 - 18x + 16 = 0$ (c) $x^2 + 18x - 16 = 0$ (d) $x^2 - 18x - 16 = 0$

YOUR PERFORMANCE

Attempted: _____

Correct: _____

Wrong: _____

Accuracy: _____

Time Taken: _____

Score: _____

Major Doubt Area: _____

Answer Key

01. (c) 02. (d) 03. (b) 04. (b) 05. (c) 06. (b) 07. (c) 08. (a) 09. (d) 10. (c)
11. (a) 12. (b) 13. (a) 14. (b) 15. (c) 16. (b) 17. (c) 18. (d) 19. (b) 20. (b)